Mathematics for Computing

Assignment 1 : Set Theory

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Assignment No.1

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Introduction

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  - Example of set: $\{a,e,i,o,u\}$
  - Example of a set: Empty set (no items): $\{ \}$ or ϕ

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  - Example of set: $\{a,e,i,o,u\}$
  - Example of a set: Empty set (no items): $\{\ \}$ or ϕ

  - Example of not a set: Set of all tall people in this room
  - Example of not a set: The collection of the best football players in the world.

Set Builder Form

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   $$Q_1 = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } x > 0 \text{ and } y > 0\}$$

Set Builder Form

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4. Write a set of all integers whose square is strictly less than 50.

$$S = \{x \mid x^2 < 50\ \forall\ x \in \mathbb{Z}\}$$

Set Builder Form

5. Translate the half open real interval $[-7, \infty)$ into formal set builder notation.

$$\mathbf{S} = \{x \mid -7 \leq x \leq \infty, \forall x \in \mathbb{R}\}$$

6.Write the set D of all positive integer divisor of 24.

$$\mathbf{D} = \{x \in \mathbb{Z} \mid kx = 24 \ \forall \ k \in \mathbb{Z}\}$$

Set Builder Form

7.Define the set **T** of all pythogorean triplets.

$$\mathbf{T} = \{x, y, z \mid \sqrt{x^2 + y^2} = z, \forall x, y, z \in \mathbb{Z}^+\}$$

8.Define set M of all natural numbers that are multiple of both 3 and 5.

$$\mathbf{M} = \{x \mid x = 3k, \forall k \in \mathbb{N} \ and \ x = 5q, \forall \ q \in \mathbb{N}\}$$

Set Builder Form

9. Express the set $\mathbf{S} = \{1, 8, 27, 64, 125, ...\}$

$$S = \{x \mid x = y^3 \ \forall \ y \in \mathbb{N}\}$$

10. Define the set **M** of all real numbers that are within a distance of 5 units from the number 12 on the number line.

$$\mathbf{M} = \{x : |x - 12| \leq 5\}$$

Example for every type of set from AI and ML Domain

- **Universal Set ($\mathcal{U}$):** The set of all elements under consideration, meaning any set A is a subset of it ($A \subseteq \mathcal{U}$). *ML Example:* All possible emails that a spam filter could ever receive in the real world.
- **Finite Set:** A set with a countable, limited number of elements, where its size (cardinality) is $|S| = n$ for some integer n. *ML Example:* The target labels in a sentiment analysis model: $\{\text{Positive}, \text{Neutral}, \text{Negative}\}$.

Example for every type of set from AI and ML Domain

- **Infinite Set:** A set with an endless number of elements, where $|S| = \infty$.
  ML Example: All the possible decimal values a neural network weight can take between 0 and 1, represented mathematically as the continuous interval $[0, 1]$.

- **Empty Set ($\emptyset$):** A set containing zero elements, meaning $|\emptyset| = 0$.
  ML Example: The overlap (intersection) between a properly split Training set (A) and Testing set (B). To avoid leakage, they share no elements: $A \cap B = \emptyset$.

Example for every type of set from AI and ML Domain

- **Singleton Set:** A set containing exactly one element, meaning $|S| = 1$, or $S = \{x\}$.
  ML Example: A batch of data containing exactly one sample, used when a model is making a single real-time prediction in production.

- **Subset ($A \subseteq B$):** Every element in A is also contained in B, defined logically as $\forall x (x \in A \implies x \in B)$.
  ML Example: The Validation dataset is a subset of the total collected historical data.

Example for every type of set from AI and ML Domain

- **Disjoint Sets:** Two sets that share no common elements whatsoever, defined as $A \cap B = \emptyset$.
  ML Example: The set of images labeled "Cat" (A) and the set labeled "Car" (B), assuming an image can only be exactly one of these.

Set Operations

There are 6 major operations on set.

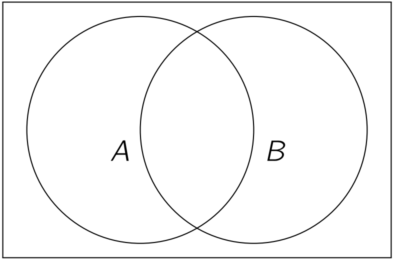
- Union of Sets
- Intersection
- Disjoint Set
- Set Difference
- Symmetric Difference
- Complement of a set

Union of Sets

The Union of Sets A and B denoted by A ∪ B, is the set of distinct elements that belong to set A or set B, or both.
The operation can be represented as:

$$A \cup B = \{x | x \in A \ or \ x \in B\}$$

The union includes every element that appears in either of the two sets, without any repetition.



$$|A \cup B \cup C|$$

Union of Sets Example

Example: Find the union of $A = \{2, 3, 4\} and B = \{3, 4, 5\}$.
Solution:

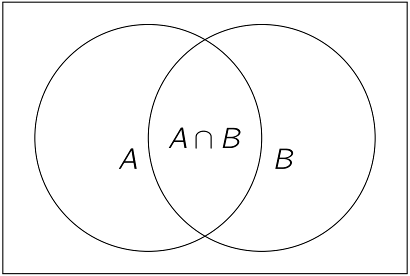
$$A \cup B = \{2, 3, 4, 5\}.$$

Intersection

The intersection of the sets A and B, denoted by A ∩ B, is the set of elements that belong to both A and B, i.e. set of the common elements in A and B. This operation is represented as:

$$A \cap B = \{x : x \in A \ and \ x \in \ B\}$$

The intersection contains only those elements that are present in both sets.



$| A \cup B |$

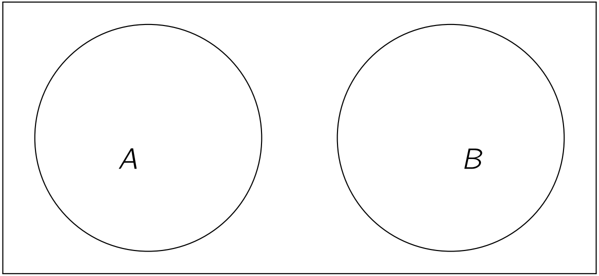
Intersection of Sets Example

Example: Find the union of $A = \{2, 3, 4\} and B = \{3, 4, 5\}$.
Solution:

$$A \cap B = \{3, 4\}.$$

Disjoint Set

Two sets are said to be disjoint if their intersection is the empty set. i.e., sets have no common elements. In simpler terms, they don't "overlap" at all. So if you try to find their intersection, you'll get the empty set, which we denote by the symbol $\phi \ or \ \{\}$.
The sets A and B are disjoint, meaning they have no common elements (no overlap).



Disjoint Sets Example

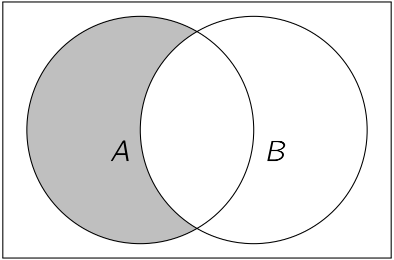
Example:The sets $A = \{2, 3, 4\} and B = \{5, 6, 7\}$. are disjoint

Example:The sets $A = \{a, e, i, o, u\}$ *and* $B = \{b, c, d\}$. are disjoint

Example:The sets $A = \{10, 100, 100, 1000\}$ *and* $B = \{a, b, c, d\}$. are disjoint

Set Difference

The difference between sets is denoted by $A - B$, which is the set containing elements that are in A but not in B i e , all elements of A except the element of B.



Set Difference Example

Example: If $A = \{1, 2, 3, 4, 5\} and B = \{2, 4, 6, 8\}, find\ A - B$.
Solution:

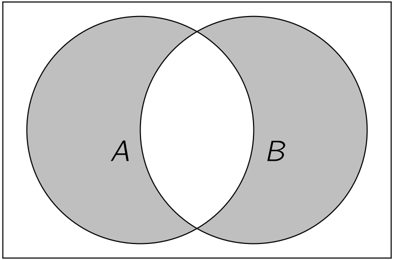
$$A - B = \{1, 3, 5\}.$$

Symmetric Difference

The symmetric difference of A and B includes elements in A or B but not both.It is denoted by

$$A \triangle B$$

$$A \triangle B = (A - B) \cup (B - A)$$



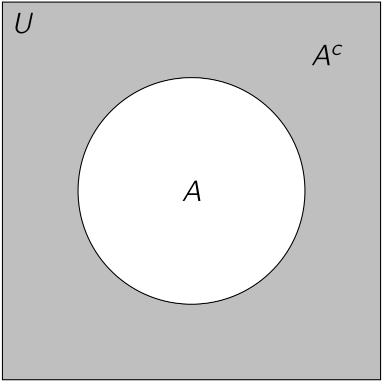
Symmetric Difference Example

Example: If $A = \{1, 2, 3\} and B = \{3, 4, 5\}$, find the symmetric difference.
Solution:

$$A \triangle B = \{1, 2, 4, 5\}.$$

Complement of a Set

If U is a universal set and X is any subset of U, then the complement of X consists of all the elements in U that are not in X.



Example of Complement of a Set

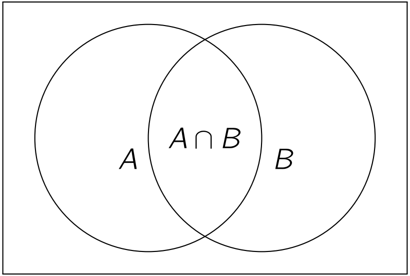
Example: If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ *and* $A = \{3, 4, 5\}$, find $A^{'}$.
Solution:

$$A^c = \{1, 2, 6, 7, 8, 9\}.$$

Inclusion - Exclusion Principle

- **What is Inclusion Exclusion Principle?**
  The Inclusion Exclusion principle is a counting method used to find the number of elements in the union of multiple sets.
- For two sets **A and B**

$$|A \cup B| = |A| + |B| - |A \cap B|$$

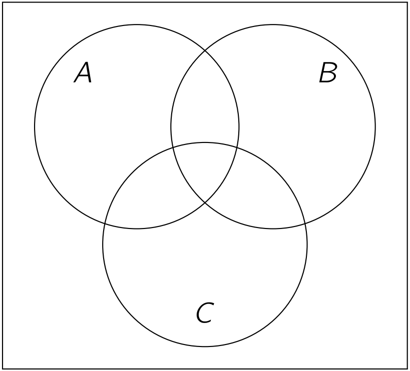


$$|A \cup B|$$

Inclusion - Exclusion Principle

- For three sets **A, B and C**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



$$|A \cup B \cup C|$$

General form of Inclusion-Exclusion Principle

For finite sets $A_1, A_2, A_3, \ldots, A_n$, the total number of elements in their union is:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|$$

General form of Inclusion-Exclusion Principle

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$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{k=1}^{n} (-1)^{k-1} \left(\sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| \right)$$

What is a Poset?

A partially ordered set is a a set where we have defined a specific rule fro how elements in a set relate to each other.

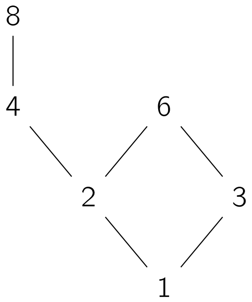
For a set to be a poset, the following three conditions must be met

- Reflexive: Every item relates to itself.
- Antisymmetric: If item **A** relates to item **B** and item **B** relates back to item **A**, then they must actaually be the exact same item.
- Transitive If **A** relates to **B** and **B** relates to **C** , then **A** automatically relates to **C**

Hasse Diagram

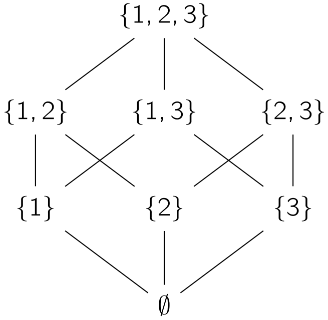
A graphical representation of a partial order relation in which all arrowheads are understood to be pointing upward is known as the Hasse Diagram.

Example: Let $A = \{1, 2, 3, 4, 6, 8\}$ be ordered by the relation "a divides b".



Hasse Diagram

Example: The power set of $S = \{1, 2, 3\}$ ordered by subset inclusion ($\subseteq$).



Thank You

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